Lecture 18

11.12.25

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Lax-Milgram

Assume that $a: V \times V \to \mathbb{R}$, $F: V \to \mathbb{R}$ satisfy

a) There exists an $\alpha \in (0,\infty)$ s.t. $|a(v_1, v_2)| \leq \alpha \|v_1\|_V \cdot \|v_2\|_V \quad \forall v_1, v_2 \in H_0^1(\Omega)$.

b) There exists a $\beta \in (0,\infty)$ s.t. $a(v,v) \geq \beta \|v\|_V^2 \quad \forall v \in H_0^1(\Omega)$.

c) There exists a $\gamma \in (0,\infty)$ s.t. $|F(v)| \leq \gamma \|v\|_V \quad \forall v \in H_0^1(\Omega)$.

Then there exists a unique solution of the variational problem (VP) and we have the estimate $\|u\|_V \leq \frac{\gamma}{\beta}$.

We focused on

$$Lu(x) = -\nabla \cdot (a(x) \nabla u(x)) + c(x) u(x) \qquad (+ \text{ Dirichlet b.c.})$$

- continuity of a: ✓ last time
- coercivity of a: we will use the Poincaré inequality (without proof)

$\forall v \in H_0^1(\Omega)$: $\|v\|^2_{L^2(\Omega)} = \int_\Omega |v(x)|^2 \, dx \leq C \int_\Omega \|\nabla v(x)\|^2 \, dx = C \|\nabla v\|^2_{L^2(\Omega)}$

where the constant C only depends on the domain

Consequence: $\|v\|_V^2 = \|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)} \leq (C+1) \int_\Omega \|\nabla v(x)\|^2 \, dx$

(but only for $v \in H_0^1(\Omega)$, otherwise false!)

We now have: $a(v,v) = \int_\Omega a(x) \nabla v(x) \cdot \nabla v(x) + c(x) v(x)^2 \, dx$

$$\geq \int_\Omega a_0 \|\nabla v(x)\|^2 + \underbrace{c_0}_{\geq 0} \underbrace{\|v(x)\|^2}_{\geq 0} \, dx$$

$$\geq \frac{a_0}{C+1} \|v\|_V^2 \qquad (\text{from the Poincaré ineq. div. by } (+1))$$

Note we do not need the Poincaré inequality if $c_0 > 0$!

$\rightarrow$ a is coercive

Conclusion: the operator

$$Lu(x) = -\nabla \cdot (a(x) \nabla u(x)) + c(x) u(x) \qquad (+ \text{ Dirichlet b.c.})$$

satisfies the assumptions required by the Theorem of Lax-Milgram and we can only solve the weak form of the PDE problem $Lu = f$.

Another example (elliptic + convection)

$$Lu(x) = -\nabla \cdot (a(x) \nabla u(x)) + \overbrace{b(x)}^{\in \mathbb{R}^d} \cdot \nabla u(x) + c(x) u(x) \qquad (+ \text{ Dirichlet})$$

$\rightarrow$ $V = H_0^1(\Omega)$

Weak formulation (exercise!):

$$a(v_1, v_2) = \int_\Omega a(x) \nabla v_1(x) \cdot \nabla v_2(x) \, dx + \underline{\tilde{b}(v_1, v_2)}$$

$$\underline{\tilde{b}(v_1, v_2)} = \int_\Omega c(x) v_1(x) v_2(x) + v_2(x) b(x) \cdot \nabla v_1(x) \, dx$$

Note: a is no longer symmetric for general b

Let us check whether Lax-Milgram applies to $b(v_1, v_2)$:

- continuity/boundedness:

$$|\tilde{b}(v_1, v_2)| \leq \int_\Omega |v_2(x)| \, \|b(x)\| \cdot \|\nabla v_1(x)\| + |c(x)| |v_1(x)| |v_2(x)| \, dx$$

$$\leq \|b\|_\infty \int_\Omega \|\nabla v_1(x)\| \cdot |v_2(x)| \, dx + \|c\|_\infty \int_\Omega |v_1(x)| \cdot |v_2(x)| \, dx$$

$$\leq (\|b\|_\infty + \|c\|_\infty) \, \|v_1\|_V \, \|v_2\|_V \qquad \checkmark$$

- coercivity:

$$\tilde{b}(v, v) = \int_\Omega c(x) v(x)^2 + v(x) \, b(x) \cdot \nabla v(x) \, dx = \int_\Omega c(x) v(x)^2 \, dx$$

$$+ \int_\Omega \frac{1}{2} \left(\nabla \cdot (b(x) v(x)^2) - v(x)^2 (\nabla \cdot b(x)) \right) dx$$

$$= \int_\Omega \left(c(x) - \frac{1}{2} \nabla \cdot b(x) \right) v(x)^2 \, dx$$

$\rightarrow$ Here we have used $\int_\Omega \frac{1}{2} \nabla \cdot (b(x) v(x)^2) \, dx = \int_\Gamma \frac{1}{2} \nu(x) \cdot (\underbrace{v(x)}_{=0 \text{ on } \Gamma}{}^2 b(x)) \, dx = 0$

$\rightarrow$ coercivity ensured if $2c(x) \geq \nabla \cdot b(x)$ for all x in Ω

For our FEM analysis we further assume:

<u>Assumption</u> (Approximation property): Let $(V_h)_{h\in(0,1)} \subset V$ be a sequence of Galerkin finite element spaces that for some $r \in \mathbb{N}$ and constant C satisfies

characterizing mesh size

$$\inf_{v_h \in V_h} \|v - v_h\|_{L^2(\Omega)} \leq C h^r \|v\|_{H^r(\Omega)}$$

and

$$\inf_{v_h \in V_h} \|v - v_h\|_{H^1(\Omega)} \leq C h^{r-1} \|v\|_{H^r(\Omega)}$$

for all $v \in H^r(\Omega)$. Note that C does not depend on $v \in H^r(\Omega)$!

Important: in particular holds true for piecewise linear elements!

<u>Theorem</u> (Lemma of Céa). Let the assumptions (Lax-Milgram + approximation property) be satisfied with $r = 2$. Assume that $u \in H^2(\Omega)$ is the exact solution of (VP) and let $u_h \in V_h$, $h \in (0,1)$, be a sequence of FEM approximations. Then

$$\|u_h - u\|_{H^1(\Omega)} \leq \tilde{C} h \|u\|_{H^2(\Omega)} \qquad \left(\text{more generally: } \|u-u_h\|_{H^1(\Omega)} \leq C h^{r-1} \|u\|_{H^r(\Omega)}\right)$$

<u>Proof</u>

a is coercive

arbitrary $v_h \in V_h$

$$\|u_h - u\|^2_{H^1(\Omega)} \leq \frac{1}{\beta} a(u_h - u, u_h - u) = \frac{1}{\beta} a(u_h - u, u_h - v_h + v_h - u)$$

a is bilinear

$$= \frac{1}{\beta} a(u_h - u, \underbrace{u_h - v_h}_{\in V_h}) + \frac{1}{\beta} a(u_h - u, v_h - u)$$

a is cont.

$$= \text{vanishes by Galerkin orthogonality} + \frac{1}{\beta} a(u_h - u, v_h - u) \leq \frac{\alpha}{\beta} \|u_h - u\|_{H^1(\Omega)} \|v_h - u\|_{H^1(\Omega)}$$

$$\Rightarrow \|u_h - u\|_{H^1(\Omega)} \leq \frac{\alpha}{\beta} \|v_h - u\|_{H^1(\Omega)} \quad \text{true for } \underline{\text{all}}\ v_h \in V_h!$$

$$\Rightarrow \|u - u_h\|_{H^1(\Omega)} \leq \frac{\alpha}{\beta} \inf_{v_h \in V_h} \|v_h - u\|_{H^1(\Omega)} \underset{\text{property}}{\overset{\text{approximation}}{\leq}} \underbrace{C \frac{\alpha}{\beta}}_{=\tilde{C}} h \|u\|_{H^2(\Omega)} \qquad \blacksquare$$